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Location of the Equivalence Point in Potentiometric Titrations

A simulated laboratory exercise

Potentiometric titration curves for acid-base precipitation, complex-formation and redox titrations, are invariably semi-logarithmic plots which are sigmoid-shaped and asymmetric. A simple method for locating the equivalence point in these titrations was described by Gunnar Gran¹ about twenty years ago. Gran's method, which has several advantages over the usual differential plot,² has been used recently for locating the equivalence point in potentiometric titrations with specific ion-electrodes³ and for a laboratory experiment in which redox titration data are evaluated with the aid of a computer.⁴ The simulated laboratory experiment described below has a two-fold purpose: to introduce the use of a small computer in the undergraduate analytical chemistry laboratory and to demonstrate the practical utility of Gran's method. The example that has been selected for the latter purpose is the titration of a diprotic acid with a strong base.

Computer Requirements

The time required for this exercise will depend on the hardware that is available. The minimum requirements are a small computer with an 8K memory, a teletype and an X-Y Plotter (e.g., the Hewlett-Packard 7002A Graphic Plotter). The addition of a high speed photoreader and punch will increase the speed of operation and improve the reliability of the system. The use of a time-shared system which could accommodate 8-10 students would allow greater flexibility and reduce the total cost per student. The programs⁵ are stored in the computer, and the necessary calculations are performed for any combination of input variables. The punched paper tape output, which has suitable scaled X-Y data, is used to drive the X-Y plotter which is off-line with the computer. The advantage of using the plotter in this manner is that the computer is free during the slowest step in the whole sequence of operations, i.e., when the X-Y plots are being made. Alternatively, the plotter can be used on-line with the computer or an X-Y oscilloscopic graphic display system can be interfaced with the computer. With the latter system, the time required for the exercise described below can be drastically curtailed and can be adapted for a lecture demonstration.

Titration Curve of a Diprotic Acid versus a Strong Base

In the titration of a diprotic acid, H_2A , versus a strong base, NaOH, it is instructive for the student to realize the effect on the titration curve when the ratio of the two acid dissociation constants of H_2A , K_1/K_2 , is varied. When K_1/K_2 is large, i.e., the two constants are well separated, the titration curve can be treated as two independent monoprotic acid titration curves. As the K_1 and K_2 values approach each other, the ratio $K_1:K_2$ becomes smaller and the two buffer regions of the two titration curves coalesce until finally, the titration curve resembles that of a monoprotic acid. The electroneutrality condition, eqn. (1), is the equation for the titration curve.

$$[Na^+] + [H^+] = [OH^-] + [HA^-] + 2[A^{2-}] \quad (1)$$

i.e.

$$\frac{C_B V_B}{(V_A + V_B)} + [H^+] = \frac{K_w}{[H^+]} + \frac{\alpha_1 C_A V_A + 2\alpha_2 C_A V_A}{(V_A + V_B)} \quad (2)$$

where V_A and V_B are the volumes in ml of the acid and NaOH added, C_A and C_B are the analytical concentrations in moles/l of the acid and the NaOH, respectively. The terms in square brackets are concentrations in moles/l and all equilibrium constants are concentration constants. It is assumed that the ionic strength remains constant during the course of the titration.

The titration curve, pH versus V_B , can be calculated from eqn. (3)

$$V_B = V_A \frac{\left(\frac{K_w}{[H^+]} + \alpha_1 C_A + 2\alpha_2 C_A - [H^+] \right)}{\left(C_B + [H^+] - \frac{K_w}{[H^+]} \right)} \quad (3)$$

where

$$\alpha_1 = \frac{K_1[H^+]}{[H^+]^2 + K_1[H^+] + K_1K_2} \quad (4)$$

and

$$\alpha_2 = \frac{K_1K_2}{[H^+]^2 + K_1[H^+] + K_1K_2} \quad (5)$$

The computer program is written for the calculation of V_B for values of $[H^+]$ starting at a specified initial value of $[H^+]$ and successively decreasing its value by a predetermined factor. In this manner, about 100 points are calculated for each titration curve. The input variables are C_A , C_B , and K_a ; V_A and K_w are read in as constants. The output is a listing of the calculated values of pH and V_B and the scaled values of pH and V_B for the X-Y plotter which can be used on-line or off-line with the computer. In the latter case, the

¹ GRAN, G., *Analyst*, **77**, 661 (1952).

² ROSSOTTI, J. F. C., AND ROSSOTTI, J., *J. CHEM. EDUC.*, **42**, 375 (1965).

³ LIBERTI, A., AND MASCINI, M., *Anal. Chem.*, **41**, 676 (1969).

⁴ MACDONALD, T. J., BARKER, B. J., AND CARUSO, J. A., *J. CHEM. EDUC.*, **49**, 200 (1972).

⁵ All programs are written in BASIC and a complete listing is made available to each student to enable him to check the equations employed and the methods used in the computations.

output consists of a punched paper tape with the scaled values of pH and V_B .

Gran's Plots

The first and second equivalence points may be located by Gran's plots that have been used for monoprotic acids.¹ Before the first equivalence point, the straight line plot of $V_B[H^+]$ versus V_B should be extrapolated to intersect the abscissa at $V_B = V_{eq1}$. After the second equivalence point, the straight line plot of $(V_A + V_B)/[H^+]$ versus V_B , upon extrapolation should intersect the abscissa at $V_B = V_{eq2}$. Although these two straight lines should be sufficient to locate both equivalence points, the equations of two more straight lines will be useful for unambiguously locating the two equivalence points. Substitution for $[A^{2-}]$ and $[HA^-]$ in the expression for K_2 gives

$$K_2 = [H^+] \cdot \frac{(V_B - V_{eq1})}{(V_{eq2} - V_B)} \quad (6)$$

After the first equivalence point, therefore, a plot of $(V_{eq2} - V_B)/[H^+]$ versus V_B should give a straight line, and before the second equivalence point, a plot of $[H^+](V_B - V_{eq1})$ versus V_B should give a straight line. It is necessary to obtain only approximate values of V_{eq1} and V_{eq2} for these two straight line plots. These approximate values can be readily determined from the straight line plots of $V_B[H^+]$ versus V_B and $(V_A + V_B)/[H^+]$ versus V_B . When the pK values of the diprotic acid are well separated ($pK_1 = 3.0$ and $pK_2 = 8.0$), there is no difficulty in locating the equivalence points and the titration curve resembles the curves obtained for two monoprotic acids (Fig. 1).⁶ When the pK values of the diprotic acid are closely spaced, e.g., $pK_1 = 5.0$ and $pK_2 = 7.0$, the titration curve resembles that of a monoprotic acid (Fig. 2) and the first equivalence point cannot be located even with Gran's plots because the straight line segments intersect above the x -axis and not on the x -axis as predicted.

For the titration curve of Na_2CO_3 versus HCl , the straight line equations for the Gran plots may be derived on the assumption that before the first equivalence point the species of importance are HCO_3^- and

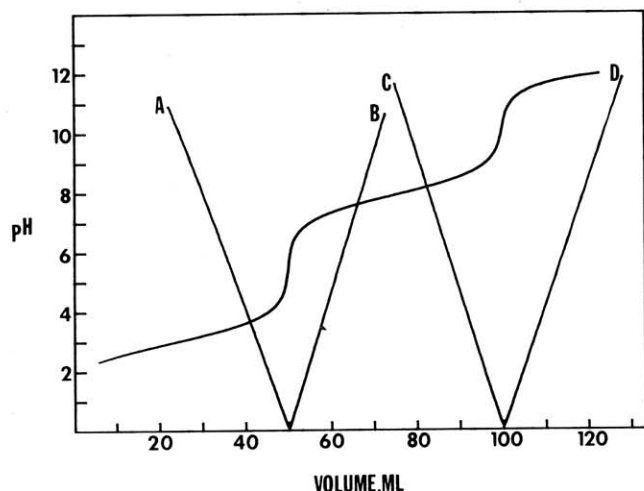


Figure 1. Titration of 50 ml of a 0.10 M solution of a diprotic acid ($pK_1 = 3.0$, $pK_2 = 8.0$) with 0.10 M NaOH. A: $V_B[H^+]$ versus V_B ; B: $(V_{eq2} - V_B)/[H^+]$ versus V_B ; C: $(V_B - V_{eq1})[H^+]$ versus V_B ; D: $(V_A + V_B)/[H^+]$ versus V_B .

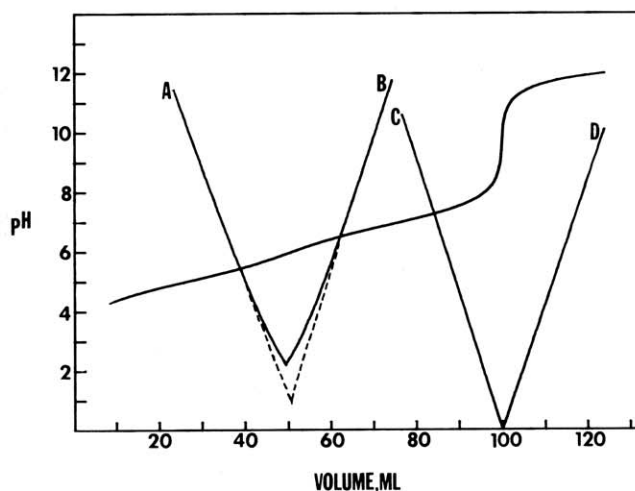


Figure 2. Titration of 50 ml of a 0.10 M solution of a diprotic acid ($pK_1 = 5.0$, $pK_2 = 7.0$) with 0.10 M NaOH. A: $V_B[H^+]$ versus V_B ; B: $(V_{eq2} - V_B)/[H^+]$ versus V_B ; C: $(V_B - V_{eq1})[H^+]$ versus V_B ; D: $(V_A + V_B)/[H^+]$ versus V_B .

CO_3^{2-} and that the addition of the strong acid, HCl , converts all the CO_3^{2-} to HCO_3^- . Before the first equivalence point

$$[CO_3^{2-}] = \frac{C_A(V_{eq1} - V_A)}{(V_A + V_B)} \quad (7)$$

and

$$[HCO_3^-] = \frac{C_A V_A}{(V_A + V_B)} \quad (8)$$

Substitution in the expression for the second dissociation constant of H_2CO_3 gives

$$K_2 = \frac{[H^+](V_{eq1} - V_A)}{V_A} \quad (9)$$

and a plot of $V_A/[H^+]$ versus V_A , before the first equivalence point should give a straight line that will intersect the x -axis at the point $(V_{eq1}, 0)$. After the first equivalence point, the concentrations of the species of importance are given by

$$[HCO_3^-] = \frac{C_A(V_{eq2} - V_A)}{(V_A + V_B)} \quad (10)$$

$$[H_2CO_3] = \frac{C_A(V_A - V_{eq1})}{(V_A + V_B)} \quad (11)$$

and substitution in the expression for K_1 , the first dissociation constant of H_2CO_3 gives

$$K_1 = \frac{[H^+](V_{eq2} - V_A)}{(V_A - V_{eq1})} \quad (12)$$

The point $(V_{eq1}, 0)$ is obtained by the intersection of the straight line plot of $[H^+](V_{eq2} - V_A)$ versus V_A with the x -axis. It is necessary to obtain only an approximate value of V_{eq2} to locate the first equivalence point exactly. If an approximate value of V_{eq1} is obtained, the second equivalence point can be located by extrapolation of the straight line plot of $(V_A - V_{eq1})/[H^+]$ versus V_A for $V_A < V_{eq2}$. When $V_A > V_{eq2}$, $[H^+]$ is given by the excess HCl added, i.e.

$$[H^+] = \frac{C_A(V_A - V_{eq2})}{(V_A + V_B)} \quad (13)$$

Extrapolation of the straight line plot $[H^+](V_A + V_B)$

⁶ In Figures 1, 2, and 3, the values of all ordinates have been multiplied by a constant for scaling purposes.

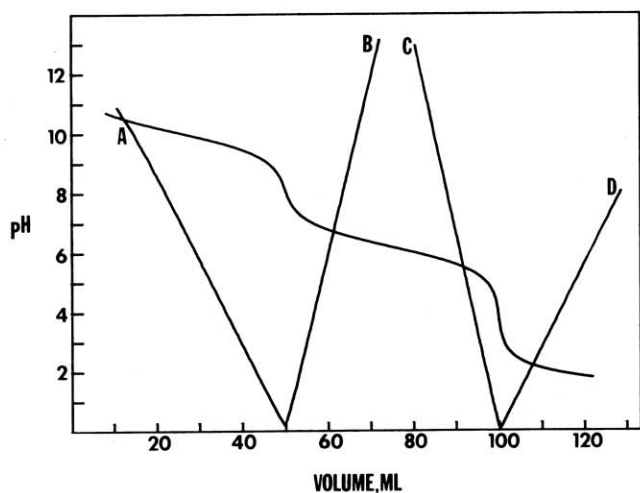


Figure 3. Titration of 50 ml of a 0.10 M solution of Na_2CO_3 ($pK_1 = 6.38$, $pK_2 = 10.32$) with 0.10 M HCl. A: $V_A/[\text{H}^+]$ versus V_A ; B: $(V_{eq2} - V_A)[\text{H}^+]$ versus V_A ; C: $(V_A - V_{eq1})/[\text{H}^+]$ versus V_A ; D: $(V_A + V_B)[\text{H}^+]$ versus V_A .

versus V_A should intersect the x -axis at the point $(V_{eq2}, 0)$. Figure 3 shows a typical curve for the titration of the Na_2CO_3 versus HCl and the Gran plots that are used for the location of the first and second equivalence points. With only minor modifications of the equations derived for the Na_2CO_3 -HCl titration, the student should be able to repeat the exercise for the titration of either a mixture of NaOH and Na_2CO_3 or NaHCO_3 and Na_2CO_3 versus HCl. His ability to make the necessary modifications in the equations as well as in the computer programs is a good index for measuring the value of this simulated laboratory exercise as a teaching tool. The student should also be made aware of the practical utility of this exercise. For example, it should be possible to detect the presence of Na_2CO_3 in a solution of NaOH ² or to determine the total alkalinity as well as the total carbonate in sea water by the use of Gran plots.⁷

⁷ DYRSSEN, D., AND SILLEN, L. G., *Tellus*, 19, 10 (1967).